

differs from the rest. If this block is identified or located by the system, it is recognised as false and is deleted from the chain.

Traceability: All the transactions that are added to a private or public blockchain are time stamped and signed digitally. This means that companies can trace every transaction back to a particular time period. And they can also locate the corresponding party on the blockchain through their public address.

DDoS: Blockchain transactions can be denied easily if the participating units are delayed from sending transactions. For example, the entire attendant infrastructure and the blockchain organisation can be crippled due to the DDoS attack on a set of entities or an entity. These kinds of attacks can introduce integrity risks to a blockchain.

Blockchain for cyber security

One interesting use case is applying the strong integrity assurance feature of blockchain technology to strengthen the cyber security of many other technologies. For example, to ensure the integrity of software downloads like firmware updates, patches, installers, etc, blockchain can be used in the same way that we make use of MD5 hashes today. Our file download that we compare against the hash might be compromised on a vendor website and altered without our knowledge. With a higher level of confidence, we can make a comparison against what is permanently recorded in the blockchain. The use of blockchain technologies has great security potential, particularly in the world of cyber-physical systems (CPS) such as IoT, industrial controls, vehicles, robotics, etc.

Summarising this, for cyber-physical systems the integrity of data is the key concern while the confidentiality in many cases is almost irrelevant. This is the key difference between cyber security for cyber-physical systems and cyber security for traditional enterprise IT systems.

Blockchain technology is just what the doctor ordered to address the key cyber-physical systems' security concerns.

The key characteristics of a blockchain that establish trust are listed below.

- **Identification and authentication:** Access is granted via cryptographic keys and access rules.
- **Transaction rules:** At every node standard rules are applied to every transaction.
- **Transaction concatenation:** Every transaction is linked to its previous transaction.
- **Consensus mechanism:** In order to agree upon the transaction integrity, mathematical puzzles are solved for all nodes.
- **Distributed ledger:** There are standards for listing transactions on every node.
- **Permissioned and unpermissioned:** Ability to participate in a blockchain can be open or pre-approved.

Is blockchain secure?

Blockchain stores data using sophisticated and innovative software rules that are extremely difficult for attackers to manipulate. The best example is Bitcoin. In Bitcoin's blockchain, the shared data is the history of every transaction made. Information is stored in multiple copies on a network of computers called nodes. Each time someone submits a transaction to the ledger, the node checks to make sure the transaction is valid or not. A subset of the package validates the transaction into blocks and adds them to the previous chain.

The blockchain offers a higher level of security to every individual user. This is because it removes the need for easily compromised and weak online identities and passwords.

How does a blockchain protect data?

Instead of uploading data to a cloud server or storing it in a single location, a blockchain breaks everything into several small nodes. A blockchain protects data because:

- It is decentralised.
- It offers encryption and validation.
- It can be private or public.
- It is virtually impossible to hack.
- It offers quality assurance.
- It ensures fast, cheap and secure transfer of funds across the globe.
- It is well known for its traceability.
- Through it, transactions become more transparent.

Cyber security is a priority, not an afterthought

It seems like in the digital age, comfort and convenience have overtaken things like privacy and security in very unfortunate ways. The handing over of personal information to companies like Facebook is a personal choice; however, no one wants to see information leaked to third parties without consent. The blockchain is all about security. It has provided us a simple, effective and affordable way of ensuring that our cyber security needs are not only met, but also exceeded. We need to understand that the technologies we use to improve our lives can also be used to harm us. That is the reality of the era we are living in, one where most of our personal data is on the cloud, on our mobile device, or on our computer. Because of that, it is vital to look at online safety and cyber security as priorities and not afterthoughts. The blockchain can assist us in turning that thought into reality, and allow us to build a future where online threats are kept to a minimum. **END** 

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